DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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TITLE- AI StoryForge – User Story Quality & Estimation Assistant

Abstract

User stories are an important part of Agile and Adaptive Software Engineering, as they describe software requirements from the perspective of users and help development teams understand the expected functionality of a system. However, poorly written, incomplete, ambiguous, or overly complex user stories can create misunderstandings between stakeholders and development teams, leading to inaccurate effort estimation, rework, delays, and reduced software quality. Manually reviewing large numbers of user stories and estimating their complexity can also be time-consuming and may depend heavily on individual team members' experience.

This project proposes AI StoryForge – User Story Quality & Estimation Assistant, an AI-integrated platform designed to analyze, improve, and estimate Agile user stories. The proposed system will support the workflow from user story input and text preprocessing to quality analysis, issue identification, improvement suggestions, and effort estimation. The system will analyze user stories based on important quality characteristics such as clarity, completeness, ambiguity, specificity, testability, and acceptance criteria. Natural Language Processing (NLP) and machine learning techniques will be used to identify potential issues in user stories and generate recommendations for improving their structure and quality.

A major focus of the project is to investigate how AI-based analysis and automated estimation can assist Agile development teams in creating better-quality user stories and making more consistent development effort estimates. The system will provide a quality score for each user story, identify missing or problematic elements, suggest an improved version, and estimate its relative complexity using Agile story points. The platform may also provide explanations for the generated score and estimation to improve transparency and user understanding. The proposed system will be evaluated based on user story quality detection, estimation accuracy, recommendation usefulness, usability, and consistency. The final platform aims to reduce manual effort in requirement analysis and support Agile teams in making faster, more informed, and data-driven decisions during software development.

Signature of the Guide HOD